ECE 332 — Homework 3: Generated Answers from Final Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

Symbol reference (used throughout):

- $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m — permittivity of free space (constant)
- $\mu_0 = 4\pi \times 10^{-7}$ H/m — permeability of free space (constant)
- ε_r — relative permittivity (dielectric constant) of the medium, dimensionless
- μ_r — relative permeability of the medium, dimensionless. $\mu_r = 1$ for nonmagnetic materials (air, most dielectrics)
- $\varepsilon = \varepsilon_r \varepsilon_0$ — absolute permittivity of the medium (F/m)
- $\mu = \mu_r \mu_0$ — absolute permeability of the medium (H/m)
- σ — conductivity of the medium (S/m). $\sigma = 0$ for lossless / ideal dielectric
- $\eta = \sqrt{\mu/\varepsilon}$ — intrinsic impedance of the medium (Ω)
- $\eta_0 = \sqrt{\mu_0/\varepsilon_0} = 120\pi \approx 377\,\Omega$ — intrinsic impedance of free space
- $k = \omega\sqrt{\mu\varepsilon} = \omega\sqrt{\varepsilon_r}/c$ (nonmag.) — wavenumber (rad/m)
- α — attenuation constant (Np/m); β — phase constant (rad/m)
- $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$ — reflection coefficient (unitless)
- $\tau = 1 + \Gamma$ — transmission coefficient (unitless)
- Superscripts on $\tilde{\mathbf{E}}$: i = incident, r = reflected, t = transmitted. Three distinct waves, three distinct superscripts.

Note: *the official HW3 solution PDF labels both reflected and transmitted as $\tilde{E}^r$ (typo). The second one is really $\tilde{E}^t$ — you can tell because it carries τ and k_2 instead of Γ and k_1. This doc uses the correct $i/r/t$ throughout.*

Problem 1 — LHC Wave Normally Incident on a Dielectric — 20 pts

Setup: 200 MHz LHC plane wave in air, $|E| = 5$ V/m, normally incident on dielectric ($\varepsilon_r = 4$) at $z \geq 0$. Positive max at $z = 0$, $t = 0$.

NORMAL INCIDENCE — Γ AND τ

Med. 1 ($z<0$): incident side. Med. 2 ($z \geq 0$): transmitted side. Boundary at $z=0$.

Reflection coeff. *wave from medium 1*

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \text{ (unitless)}$$

Transmission coeff. *always*

$$\tau = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Intrinsic impedance *general → nonmag. shortcut*

$$\eta_i = \sqrt{\tfrac{\mu_r}{\varepsilon_r}}\,\eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\varepsilon_{r,i}}}\ (\Omega) \qquad \eta_0 = 120\pi \approx 377\,\Omega$$

Default $\mu_r = 1$ (air, dielectric, "nonmag."). Only use $\mu_r \neq 1$ if problem says so ("ferromagnetic", iron, ferrite, or μ_r given explicitly).

Low-loss η_2 (poor conductor) $\sigma/(\omega\varepsilon) \ll 1$

$$\eta_2 \approx \sqrt{\tfrac{\mu}{\varepsilon'}}\left(1 + j\,\tfrac{\sigma}{2\omega\varepsilon'}\right) \xrightarrow{\text{drop tiny } j} \frac{\eta_0}{\sqrt{\varepsilon_{r,2}}}\ (\Omega)$$

For Γ/τ just use $\eta_0/\sqrt{\varepsilon_{r,2}}$. The j term is for accuracy with $|\eta_c|, \theta_\eta$.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100\,|\Gamma|^2$$

% transmitted *η-ratio matters!*

$$\%P_t = 100\,|\tau|^2\,\frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

If medium 2 is denser ($\eta_2 < \eta_1$), $|\tau| > 1$ but $\%P_t < 100\%$ because of the η_1/η_2 factor.

$\tilde{\mathbf{E}}$ PHASOR TEMPLATES

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\varepsilon_{r,i}}}{c}\ \text{(rad/m)} \quad \Rightarrow \quad k_2 = k_1\sqrt{\varepsilon_{r,2}/\varepsilon_{r,1}}$$

Direction table *$u \in \{x, y, z\}$ = axis perp. to boundary*

Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

$+\hat{u}$ direction ⇔ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$\tilde{\mathbf{E}}^i = a_0\hat{e}\,e^{\mp jk_1u}$, $\tilde{\mathbf{E}}^r = \Gamma a_0\hat{e}\,e^{\pm jk_1u}$, $\tilde{\mathbf{E}}^t = \tau a_0\hat{e}\,e^{\mp j}$

(lossy med 2: replace $e^{\mp jk_2u}$ with $e^{\mp\alpha_2u}e^{\mp j\beta_2u}$)

Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x}+\hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x\hat{x} + a_y e^{j\delta}\hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

Wave parameters in medium 1 (air):

Angular frequency from f:

$$\omega = 2\pi f = 2\pi(2\times10^8) = 4\pi\times10^8 \text{ rad/s}$$

Wavenumber from the lossless plane wave formula $k = \omega\sqrt{\varepsilon_r}/c$. In air, $\varepsilon_r = 1$ so this reduces to $k_1 = \omega/c$:

$$k_1 = \tfrac{\omega}{c} = \tfrac{4\pi\times10^8}{3\times10^8} = \tfrac{4\pi}{3} \text{ rad/m}$$

(a) LHC incident phasor. The PHASOR TEMPLATES section says the incident wave has the form $\tilde{\mathbf{E}}^i = a_0\hat{e}\,e^{-jk_1z}$ where $\hat{e}$ depends on polarization. From the polarization table:

$$\text{LHC} \Rightarrow \hat{e} = \hat{x} + j\hat{y}$$

With $a_0 = 5$ V/m (the modulus) and $k_1 = 4\pi/3$:

$$\boxed{\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y})\,e^{-j4\pi z/3} \text{ V/m}}$$

(b) Reflection coefficient Γ and transmission coefficient τ.

General intrinsic impedance: $\eta = \sqrt{\mu/\varepsilon}$.

Decompose into relative + free-space constants: since $\mu = \mu_r\mu_0$ and $\varepsilon = \varepsilon_r\varepsilon_0$,

$$\eta = \sqrt{\frac{\mu_r\mu_0}{\varepsilon_r\varepsilon_0}} = \sqrt{\frac{\mu_r}{\varepsilon_r}} \cdot \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{\frac{\mu_r}{\varepsilon_r}}\,\eta_0$$

which for nonmagnetic media ($\mu_r = 1$) reduces to $\eta = \eta_0/\sqrt{\varepsilon_r}$.

Compute η_0 from μ_0, ε_0 directly:

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{(4\pi\times10^{-7})(36\pi\times10^{9})} = 120\pi\,\Omega \approx 377\,\Omega$$

Medium 1 (air): $\mu_r = 1, \varepsilon_r = 1$. So $\eta_1 = \eta_0\sqrt{1/1} = \eta_0 = 120\pi\,\Omega$.

Medium 2 (nonmagnetic dielectric, $\mu_r = 1, \varepsilon_r = 4$): use the shortcut $\eta = \eta_0/\sqrt{\varepsilon_r}$:

$$\eta_2 = \frac{\eta_0}{\sqrt{\varepsilon_{r2}}} = \frac{120\pi}{\sqrt{4}} = \frac{120\pi}{2} = 60\pi\,\Omega$$

Now apply the boundary formula $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$:

$$\Gamma = \frac{60\pi - 120\pi}{60\pi + 120\pi} = \frac{-60\pi}{180\pi} = \boxed{-\frac{1}{3}}$$

$$\tau = 1 + \Gamma = 1 - \frac{1}{3} = \boxed{\frac{2}{3}}$$

(c) Reflected, transmitted, and total field phasors.

Wavenumbers in both media use the same formula $k_i = \omega\sqrt{\varepsilon_{r,i}}/c$, just plug in each medium's ε_r:

Medium 1 (air, $\varepsilon_{r,1} = 1$):

$$k_1 = \frac{\omega\sqrt{1}}{c} = \frac{4\pi\times10^8}{3\times10^8} = \frac{4\pi}{3}\ \text{rad/m}$$

Medium 2 (dielectric, $\varepsilon_{r,2} = 4$):

$$k_2 = \frac{\omega\sqrt{4}}{c} = \frac{2\omega}{c} = 2k_1 = \frac{8\pi}{3}\ \text{rad/m}$$

Notice: same formula, only ε_r changes per medium (just like η).

From the PHASOR TEMPLATES (reflected wave travels in $-\hat{z}$, sign of z-term flips):

$$\tilde{\mathbf{E}}^r = \Gamma \cdot a_0\hat{e}\,e^{+jk_1z} = -\frac{1}{3}\cdot 5(\hat{x} + j\hat{y})\,e^{+j4\pi z/3} = -\frac{5}{3}(\hat{x} + j\hat{y})\,e^{+j4\pi z/3}$$

Transmitted wave in lossless medium 2:

$$\tilde{\mathbf{E}}^t = \tau \cdot a_0\hat{e}\,e^{-jk_2z} = \frac{2}{3}\cdot 5(\hat{x} + j\hat{y})\,e^{-j8\pi z/3} = \frac{10}{3}(\hat{x} + j\hat{y})\,e^{-j8\pi z/3}$$

Total field in medium 1 is incident + reflected:

$$\boxed{\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r = 5(\hat{x} + j\hat{y})\left(e^{-j4\pi z/3} - \frac{1}{3}e^{+j4\pi z/3}\right)\ \text{V/m}}$$

(d) Percentages of incident average power reflected and transmitted.

From the POWER REFLECTION snippet, $\%P_r = 100|\Gamma|^2$ and $\%P_t = 100|\tau|^2(\eta_1/\eta_2)$. The η_1/η_2 factor accounts for the change in wave impedance: even though $|\tau| > 0$, the medium beyond carries power differently.

$$\%P_r = 100\cdot\left|-\frac{1}{3}\right|^2 = 100\cdot\frac{1}{9} = \boxed{11.1\%}$$

$$\%P_t = 100\cdot\left|\frac{2}{3}\right|^2\cdot\frac{120\pi}{60\pi} = 100\cdot\frac{4}{9}\cdot 2 = \boxed{88.9\%}$$

Sanity check: $11.1\% + 88.9\% = 100\%$. Energy conserved.

Problem 2 — Same Wave, Poor-Conductor Medium 2 20 pts

Setup: Replace medium 2 with poor conductor: $\varepsilon_r = 2.25$, $\mu_r = 1$, $\sigma = 10^{-4}$ S/m. Everything else same as P1.

NORMAL INCIDENCE — Γ AND τ

Med. 1 *($z < 0$): incident side.* ***Med. 2*** *($z \geq 0$): transmitted side. Boundary at $z = 0$.*

Reflection coeff. *wave from medium 1*

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \text{ (unitless)}$$

Transmission coeff. *always*

$$\tau = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Intrinsic impedance *general → nonmag. shortcut*

$$\eta_i = \sqrt{\frac{\mu_r}{\varepsilon_r}}\,\eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\varepsilon_{r,i}}}\ (\Omega) \qquad \eta_0 = 120\pi \approx 377\,\Omega$$

Default $\mu_r = 1$ (air, dielectric, "nonmag."). Only use $\mu_r \neq 1$ if problem says so ("ferromagnetic", iron, ferrite, or μ_r given explicitly).

Low-loss η_2 (poor conductor) *$\sigma/(\omega\varepsilon) \ll 1$*

$$\eta_2 \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\sigma}{2\omega\varepsilon'}\right) \xrightarrow{\text{drop tiny } j} \frac{\eta_0}{\sqrt{\varepsilon_{r,2}}}\ (\Omega)$$

For Γ/τ just use $\eta_0/\sqrt{\varepsilon_{r,2}}$. The j term is for accuracy with $|\eta_c|, \theta_\eta$.

LOSSY MEDIA — 5 CASES

Loss tangent *$\varepsilon' = \varepsilon_r\varepsilon_0$, $\varepsilon_0 \approx 1/(36\pi \times 10^9)$*

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

Type	$\sigma/\omega\varepsilon$	η_c, α, β
Lossless ($\sigma = 0$)	0	$\eta = \eta_0/\sqrt{\varepsilon_r}$ exact; $\alpha = 0$, $\beta = \omega\sqrt{\varepsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\varepsilon_r}}\left(1+j\frac{\sigma}{2\omega\varepsilon'}\right)$; $\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}$, $\beta \approx \frac{\omega\sqrt{\varepsilon_r}}{c}$
Quasi-cond.	10^{-2}–10^2	exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$, see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f\mu/\sigma}$; $\alpha \approx \beta \approx \sqrt{\pi f\mu\sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\varepsilon_r}\,\eta_0$

For Γ/τ: drop j correction; use $\eta_0/\sqrt{\varepsilon_r}$. Magnetic: keep μ_r.

Exact (quasi-cond) $X = \sqrt{1 + (\varepsilon''/\varepsilon')^2}$

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X-1},\ \beta = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X+1},\ \theta_\eta = \frac{1}{2}\arctan$$

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100\,|\Gamma|^2$$

% transmitted *η-ratio matters!*

$$\%P_t = 100\,|\tau|^2\,\frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

If medium 2 is denser ($\eta_2 < \eta_1$), $|\tau| > 1$ but $\%P_t < 100\%$ because of the η_1/η_2 factor.

Wave parameters in medium 1 (air). Same as P1, since medium 1 hasn't changed:

$$\omega = 2\pi f = 4\pi \times 10^8 \text{ rad/s}, \quad k_1 = \frac{\omega\sqrt{\varepsilon_{r,1}}}{c} = \frac{\omega}{c} = \frac{4\pi}{3} \text{ rad/m}, \quad \eta_1 = \eta_0 = 120\pi\,\Omega$$

(a) LHC incident phasor. Identical to P1 — incident wave is in medium 1 (air), which didn't change. From the PHASOR TEMPLATES + polarization table ($\hat{e}_{\text{LHC}} = \hat{x} + j\hat{y}$):

$$\boxed{\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y})\,e^{-j4\pi z/3} \text{ V/m}}$$

(b) Reflection and transmission coefficients — but first classify medium 2.

Compute the loss tangent $\sigma/(\omega\varepsilon)$ to pick which formulas to use:

$$\frac{\sigma_2}{\omega\varepsilon_2} = \frac{\sigma_2}{\omega\varepsilon_{r,2}\varepsilon_0} = \frac{10^{-4}}{(4\pi \times 10^8)(2.25)(8.854 \times 10^{-12})} = 4 \times 10^{-3}$$

This is $\ll 1$, so medium 2 is a **low-loss dielectric**. The LOSSY MEDIA — CLASSIFICATION table tells us to use the LOW-LOSS QUICK PARAMS shortcuts (not the full exact lossy formulas).

Intrinsic impedance — same derivation chain as P1, with the low-loss correction. From the cheat sheet's NORMAL INCIDENCE ebox the general formula is $\eta = \sqrt{\mu_r/\varepsilon_r}\,\eta_0$. For low-loss media the LOW-LOSS QUICK PARAMS box says:

$$\eta_2 \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\sigma}{2\omega\varepsilon'}\right) \xrightarrow[\sigma/\omega\varepsilon \ll 1]{\mu_r=1} \frac{\eta_0}{\sqrt{\varepsilon_r}}$$

The j correction is $4\times10^{-3}/2 = 2\times10^{-3}$, tiny enough to drop for Γ, τ computation. So:

$$\eta_2 \approx \frac{\eta_0}{\sqrt{\varepsilon_{r,2}}} = \frac{120\pi}{\sqrt{2.25}} = \frac{120\pi}{1.5} = 80\pi\,\Omega$$

Attenuation and phase constants from LOW-LOSS QUICK PARAMS (these matter for the transmitted-wave decay even though they don't affect Γ):

$$\alpha_2 \approx \frac{\sigma_2\eta_0}{2\sqrt{\varepsilon_{r,2}}} = \frac{10^{-4}\cdot120\pi}{2\sqrt{2.25}} = 1.26\times10^{-2}\ \text{Np/m}$$

$$\beta_2 \approx \frac{\omega\sqrt{\varepsilon_{r,2}}}{c} = \frac{(4\pi\times10^8)(1.5)}{3\times10^8} = 2\pi\ \text{rad/m}$$

Apply the boundary formulas:

$$\Gamma = \frac{\eta_2-\eta_1}{\eta_2+\eta_1} = \frac{80\pi-120\pi}{80\pi+120\pi} = \frac{-40\pi}{200\pi} = \boxed{-\frac{1}{5}}$$

$$\tau = 1+\Gamma = 1-\frac{1}{5} = \boxed{\frac{4}{5}}$$

(c) Reflected, transmitted, and total field phasors.

Direction signs from the DIRECTION TABLE. Med 2 is at $z \geq 0$ (same as P1), so use the first row: incident gets $-jk_1z$, reflected gets $+jk_1z$, transmitted gets the lossy decay $e^{-\alpha_2 z}e^{-j\beta_2 z}$ (since med 2 is now lossy, replace e^{-jk_2z} with $e^{-\alpha_2 z}e^{-j\beta_2 z}$).

Reflected template (still in lossless medium 1, traveling $-\hat{z}$):

$$\tilde{\mathbf{E}}^r = \Gamma\, a_0\hat{e}\, e^{+jk_1z}$$

Plug in $\Gamma = -1/5$, $a_0 = 5$, $\hat{e} = \hat{x}+j\hat{y}$, $k_1 = 4\pi/3$:

$$\tilde{\mathbf{E}}^r = -\frac{1}{5}\cdot 5(\hat{x}+j\hat{y})\,e^{+j4\pi z/3} = -(\hat{x}+j\hat{y})\,e^{+j4\pi z/3}\ \text{V/m}$$

Transmitted template (lossy medium 2, traveling $+\hat{z}$, replace e^{-jk_2z} with $e^{-\alpha_2 z}e^{-j\beta_2 z}$):

$$\tilde{\mathbf{E}}^t = \tau\, a_0\hat{e}\, e^{-\alpha_2 z}\, e^{-j\beta_2 z}$$

Plug in $\tau = 4/5$, $a_0 = 5$, $\hat{e} = \hat{x}+j\hat{y}$, $\alpha_2 = 1.26\times10^{-2}$, $\beta_2 = 2\pi$:

$$\tilde{\mathbf{E}}^t = \frac{4}{5}\cdot 5(\hat{x}+j\hat{y})\,e^{-\alpha_2 z}e^{-j\beta_2 z} = 4(\hat{x}+j\hat{y})\,e^{-1.26\times10^{-2}z}\,e^{-j2\pi z}\ \text{V/m}$$

Total field in medium 1 ($z \leq 0$):

$$\boxed{\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r = 5(\hat{x}+j\hat{y})\Big(e^{-j4\pi z/3} - \frac{1}{5}e^{+j4\pi z/3}\Big)\ \text{V/m}}$$

(d) Percentages of incident average power reflected and transmitted.

From POWER REFLECTION snippet, same formulas as P1:

$$\%P_r = 100|\Gamma|^2 = 100\cdot\left|-\frac{1}{5}\right|^2 = 100\cdot\frac{1}{25} = \boxed{4\%}$$

$$\%P_t = 100|\tau|^2\frac{\eta_1}{\eta_2} = 100\cdot\left|\frac{4}{5}\right|^2\cdot\frac{120\pi}{80\pi} = 100\cdot\frac{16}{25}\cdot\frac{3}{2} = \boxed{96\%}$$

Sanity check: $4\%+96\% = 100\%$. Less reflection than P1 (4% vs 11.1%) because medium 2 is closer in impedance to air ($\eta_2 = 80\pi$ vs 60π); the smaller impedance jump means more of the wave gets through.

Problem 3 — Parallel-Polarized Wave Through Two Dielectric Layers 10 pts

Setup: $\theta_i = 30°$ from air, layer 1 ($\varepsilon_r = 6.25$, $t = 5$ cm), layer 2 ($\varepsilon_r = 2.25$, $t = 5$ cm), exits to air.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin\theta_2 = \sin\theta_1 \sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}} = \sin\theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs* $1 \to 2 \to 3 \to 4$

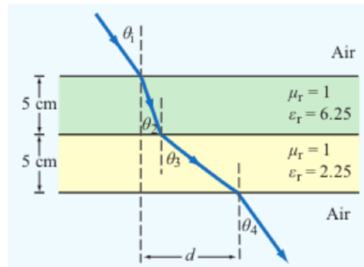
$$\sin\theta_{i+1} = \sin\theta_i \sqrt{\frac{\varepsilon_{r,i}}{\varepsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes *need Fresnel coefficients (parallel vs perpendicular).*

LATERAL DISPLACEMENT



Beam offset through N slabs *thickness t_i, refracted angle inside slab i*

Slab-indexed *θ_i = angle inside slab i*

$$d = \sum_{i=1}^{N} t_i \tan\theta_i$$

Angle-indexed (HW3 P3 style) *θ_1 = incident, θ_{i+1} = inside slab i*

$$d = \sum_{i=1}^{N} t_i \tan\theta_{i+1}$$

Same physics, different numbering. In HW3 P3: slab 1 contributes $t_1 \tan\theta_2$, slab 2 contributes $t_2 \tan\theta_3$. The angle in the formula is always the one the ray makes inside *that slab.*

Air–slab–air: exit angle returns to θ_1 but is laterally shifted by d.

(a) Chained Snell's law:

$$\sin\theta_2 = \sin 30° \sqrt{\frac{1}{6.25}} = 0.5 \cdot 0.4 = 0.2 \Rightarrow \theta_2 = \sin^{-1}(0.2) = \boxed{11.54°}$$

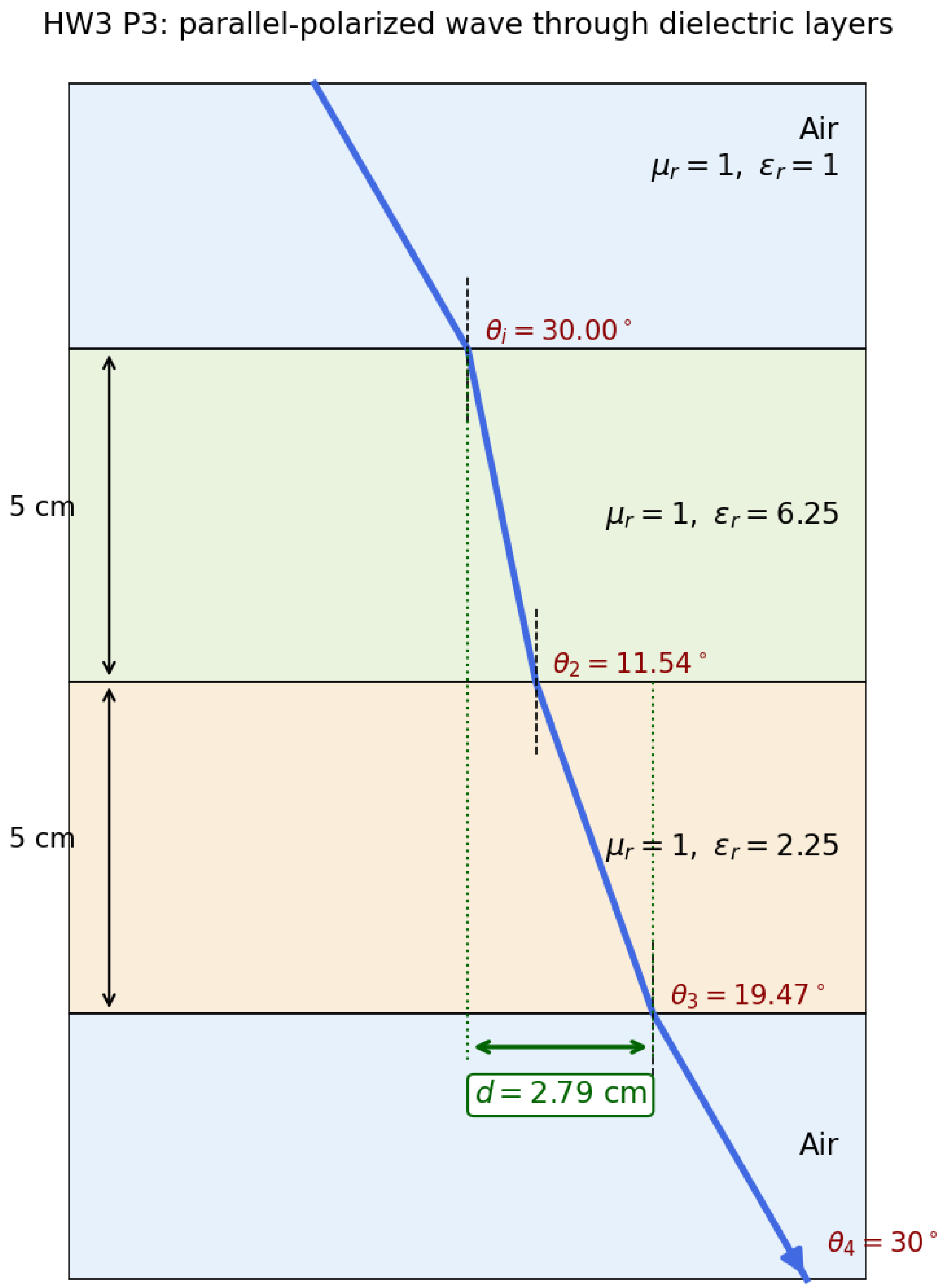
$$\sin\theta_3 = 0.2\sqrt{\frac{6.25}{2.25}} = 0.2 \cdot \frac{5}{3} = \frac{1}{3} \Rightarrow \theta_3 = \sin^{-1}(1/3) = \boxed{19.47°}$$

$$\sin\theta_4 = \frac{1}{3}\sqrt{\frac{2.25}{1}} = \frac{1}{3} \cdot \frac{3}{2} = \frac{1}{2} \Rightarrow \theta_4 = \boxed{30° = \theta_i}$$

(b) Lateral displacement:

$$d = t_1 \tan\theta_2 + t_2 \tan\theta_3 = 5\tan 11.54° + 5\tan 19.47° = 5(0.2041 + 0.3536)\ \text{cm}$$

$$\boxed{d = 2.79\ \text{cm}}$$


HW3 P3: parallel-polarized wave through dielectric layers
Air
$\mu_r = 1,\ \varepsilon_r = 1$
$\theta_i = 30.00^\circ$
5 cm
$\mu_r = 1,\ \varepsilon_r = 6.25$
$\theta_2 = 11.54^\circ$
5 cm
$\mu_r = 1,\ \varepsilon_r = 2.25$
$\theta_3 = 19.47^\circ$
$d = 2.79$ cm
Air
$\theta_4 = 30^\circ$

Problem 4 — Dielectric Waveguide TE Mode Identification 20 pts

Setup: TE wave, $a = 5$ cm, $b = 3$ cm, $E_x = -36\cos(40\pi x)\sin(100\pi y)\sin\big((2.4\pi\times10^{10})t - 52.9\pi z\big)$ V/m.

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0,\ H_z \neq 0$
TM mode $H_z = 0,\ E_z \neq 0$
Dimensions $a \times b$ *with* $a > b$ *along* $\hat{x}, \hat{y}$*; wave in* $\hat{z}$.
E_x **form (TE)** *read* m, n *from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)\sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$*; coef. on* $y = n\pi/b$.

Identify TE$_{mn}$ vs TM$_{mn}$ *cos/sin pattern of* E_x *is identical for both!*
1. Read the problem statement (e.g. "a TE wave" → TE).
2. If $m = 0$ or $n = 0$ in the mode label ⇒ must be TE (TM forbids 0).
3. Source field $H_z(x, y)$ given ⇒ TE. Source field $E_z(x, y)$ given ⇒ TM.

TE allows at least one of $m, n \geq 1$ *(other can be 0). TM needs both* $m, n \geq 1$*. That's why* TE$_{10}$ *exists but* TM$_{10}$ *doesn't.*

CUTOFF FREQUENCY

f_{mn} *m, n integers ≥ 0 (not both zero)*

$$f_{mn} = \frac{u_{p0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}\ \text{(Hz)}$$

$$u_{p0} = \frac{c}{\sqrt{\varepsilon_r}}\ \text{(m/s, guide medium)}$$

Common cutoffs $(a > b)$

Mode	f_c
TE$_{10}$	$u_{p0}/(2a)$ (dominant)
TE$_{01}$	$u_{p0}/(2b)$
TE$_{20}$	$u_{p0}/a = 2f_{10}$
TE$_{11}$,TM$_{11}$	$\frac{u_{p0}}{2}\sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$.

ε_r FROM DISPERSION

Given ω, β, **mode** (m, n)

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]$$

Sanity: β *in rad/m,* a, b *in m,* ω *in rad/s.*

WAVE IMPEDANCE & H_y

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}}\ (\Omega) \qquad Z_{\text{TM}} = \eta\sqrt{1 - (f_c/f)^2}$$

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} \qquad H_y = \frac{E_x}{Z_{\text{TE}}}\ \text{(TE mode)}$$

$Z_{TE} > \eta$ *always (denominator* < 1*);* $Z_{TM} < \eta$.

(a) Mode from the field expression. Read $m\pi/a$ from the cos coefficient and $n\pi/b$ from the sin coefficient.

$$\frac{m\pi}{a} = 40\pi \Rightarrow m = 40\cdot 0.05 = 2$$

$$\frac{n\pi}{b} = 100\pi \Rightarrow n = 100\cdot 0.03 = 3$$

$$\boxed{\text{Mode: TE}_{23}}$$

(b) ε_r from dispersion, with $\omega = 2.4\pi\times10^{10}$ rad/s, $\beta = 52.9\pi$ rad/m:

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right] = \left(\frac{3\times10^8}{2.4\pi\times10^{10}}\right)^2\left[(52.9\pi)^2 + (40\pi)^2 + (100\pi)^2\right]$$

$$\boxed{\varepsilon_r = 2.25}$$

(c) Cutoff frequency for TE$_{23}$:

$$u_{p0} = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3\times10^8}{1.5} = 2\times10^8\ \text{m/s}$$

$$f_{23} = \frac{u_{p0}}{2}\sqrt{\left(\frac{2}{a}\right)^2 + \left(\frac{3}{b}\right)^2} = 10^8\sqrt{(2/0.05)^2 + (3/0.03)^2} = 10^8\sqrt{1600 + 10000}$$

$$\boxed{f_{23} = 10.77 \text{ GHz}}$$

(d) H_y from $Z_{\mathbf{TE}}$: operating frequency $f = \omega/(2\pi) = 12$ GHz.

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1-(f_c/f)^2}} = \frac{377/\sqrt{2.25}}{\sqrt{1-(10.77/12)^2}} = \frac{251.3}{\sqrt{1-0.805}} = \frac{251.3}{0.442} = 569.9\ \Omega$$

$$\boxed{H_y = \frac{E_x}{Z_{\text{TE}}} = -0.063\cos(40\pi x)\sin(100\pi y)\sin((2.4\pi\times10^{10})t - 52.9\pi z)\ \text{A/m}}$$

Problem 5 — Hollow Guide, Multi-Mode Excitation & Travel Times 8 pts

Setup: Hollow guide ($\varepsilon_r = 1$, $u_{p0} = c$), $a = 3$ cm, $b = 2$ cm. Pulse with 9.5 GHz carrier, guide length 100 m.

CUTOFF FREQUENCY

f_{mn} *m, n integers ≥ 0 (not both zero)*

$$f_{mn} = \frac{u_{p0}}{2}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

$$u_{p0} = \frac{c}{\sqrt{\varepsilon_r}} \text{ (m/s, guide medium)}$$

Common cutoffs ($a > b$)

Mode	f_c
TE_{10}	$u_{p0}/(2a)$ (dominant)
TE_{01}	$u_{p0}/(2b)$
TE_{20}	$u_{p0}/a = 2f_{10}$
TE_{11},TM_{11}	$\frac{u_{p0}}{2}\sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0}\sqrt{1-(f_{mn}/f)^2} \text{ (m/s)}$$

$$u_p = \frac{u_{p0}}{\sqrt{1-(f_{mn}/f)^2}} \quad \Rightarrow \quad u_p u_g = u_{p0}^2$$

$t = L/u_g$ (s) travel time through guide of length L

Higher modes have lower u_g $\Rightarrow$ they arrive later. Group delay spread = sum over excited modes.

Cutoff frequencies for the lowest modes:

$$f_{10} = \frac{c}{2a} = \frac{3\times10^8}{0.06} = 5 \text{ GHz}$$

$$f_{01} = \frac{c}{2b} = \frac{3\times10^8}{0.04} = 7.5 \text{ GHz}$$

$$f_{11} = \frac{c}{2}\sqrt{\frac{1}{a^2}+\frac{1}{b^2}} = (1.5\times10^8)\sqrt{(1/0.03)^2+(1/0.02)^2} = 9.01 \text{ GHz}$$

$$f_{20} = \frac{c}{a} = 10 \text{ GHz} \quad (> 9.5, \text{ not excited})$$

Modes excited (carrier ¿ f_c): TE_{10}, TE_{01}, TE_{11}, TM_{11}. TE_{20} is just above cutoff so it does not propagate.

Group velocities: $u_g = c\sqrt{1-(f_{mn}/f)^2}$ with $f = 9.5$ GHz.

Mode	f_c (GHz)	u_g/c	u_g (m/s)
TE_{10}	5.00	$\sqrt{1-(5/9.5)^2} = 0.85$	2.55×10^8
TE_{01}	7.50	$\sqrt{1-(7.5/9.5)^2} = 0.61$	1.84×10^8
TE_{11}, TM_{11}	9.01	$\sqrt{1-(9.01/9.5)^2} = 0.32$	0.95×10^8

Travel times: $t = L/u_g$ **over 100 m.**

Mode	t (μs)
TE_{10}	$\boxed{0.39}$
TE_{01}	$\boxed{0.54}$
TE_{11}, TM_{11}	$\boxed{1.05}$

Higher modes arrive later because they propagate closer to cutoff, so u_g drops.